Mehdi Aminazadeh

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Master's student in Computer Science with hands-on experience in developing and deploying intelligent solutions to real-world problems, with a focus on machine learning, deep learning, and optimization

Experience

Fraunhofer IOSB – Working Student, Data Engineer Jan 2026 – Present

- Performed image segmentation and image processing for drone-detection applications such as MODEAS and ArGUS, using OpenCV, Pillow, and scikit-image.
- Implemented an image-enhancement pipeline for batches of 100 infrared / thermal images of unmanned aerial systems using ProcessPoolExecutor.
- Improved image-transformation throughput by 20% compared with a sequential one-image-at-a-time workflow, resulting in better CPU utilization.

Department of Mechanical & Process Engineering (RPTU) Mar 2026 - Present
Research Assistant

- Developed optimization workflows for factory layout planning using D-Wave Ocean SDK, QUBO/BQM formulations, and hybrid quantum-classical solvers.
- Increased feasibility rate on environmental datasets(Factory/ Vehicles) from 63% to 81% by tuning penalty coefficients, variable encodings, feature selection, and dwave-hybrid pipeline using dimod, dwave-system and Optuna.

Department of Economics (RPTU) Jan 2026 – Mar 2026
Associate Programmer

- Designed and implemented an automated data-collection pipeline for economics articles, working papers, and metadata using Crossref, APIs, DOM parsing, and Selenium.
- Built an NLP-based matching system to identify corresponding article–working paper pairs using SBERT, TF-IDF with cosine similarity, and BLEU/ROUGE.
- Enhanced document-linking accuracy and reduced manual review through tokenization, semantic similarity, and metadata-based filtering.

Institute of Machine Elements, Gears and Transmissions (RPTU) Sep 2025 – Jan 2026
Associate Programmer

- Preprocessed raw experimental FVA3 oil-lubrication data for roller bearings into a structured dataset with 490 rows and 20 features.
- Implemented and evaluated classical machine learning models as well as deep learning approaches, including Graph Neural Networks, TabNet, and FT-Transformer.
- Improved prediction accuracy for lubricant-formula performance from 91% to 94% using tabular neural network models.

Traffic Department, Isfahan Municipality Aug 2023 – Sep 2024

- Developed a traffic data pipeline and dashboard using SCATS and city climate data.
- Applied clustering and time-series analysis to identify traffic patterns and improve traffic control.

Skills & Expertise

Programming Languages: Python – SQL – PostgreSQL – Apache Spark – Apache Hadoop

Expertise: Optimization – Image Processing – Deep Learning – Pattern Recognition – NLP

Libraries: TensorFlow / Keras – Tokenizers – PyTorch / Lightning – Scikit-learn – XGBoost – Streamlit – Flask

Tools & Software: Docker – AWS – Git – Power BI

Soft Skills: Problem solving, Familiar with Agile methodologies, Fast learner

Projects

Oil Well Damage Detection – SEG-Y Data Conversion Tool **Apr 2024 – Sep 2024**

- Developed a Python tool using ObsPy, NumPy, and Pandas to convert oil well sensor data (0–120 radiation values) from CSV to SEG-Y format.
- Automated the detection of damaged borehole sections by analyzing reduced reflection values as indicators of structural weaknesses.

Device Access Anomaly Detection (IoT Security) **May 2023 – Aug 2023**

- Built a local IoT system for capturing and logging device login activity using Flask.
- Developed a detector for anomalous login attempts using machine learning methods, including Isolation Forest and rule-based approaches.

Education

Technical University of Kaiserslautern (RPTU) **Oct 2024 – Present**

M.Sc. Computer Science

University of Isfahan PNU, Iran **Sep 2018 – Jul 2022**

B.Sc. Computer Engineering
Project : Risk analysis of Macroeconomic factors using Random forest and Data scraping

Interests

- Deep neural networks
- CNNs, RNNs (GRU)
- Natural Language Processing (NLP) – Seq2Seq networks
- Graph neural networks
- MLOps – End-to-End Deep Learning
- Time-series analysis
- Pattern recognition

Language

- **English**: C1 (Academic IELTS 7.0 / 9.0) | TRF Number: 23IR005521AMIM120A
- **German**: B1

Hobbies

Swimming, Basketball, Reading, Music, Gaming(Dota 2)